Project Development Phase

Dashboard and Story

Project Title

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

Dashboard

Connect Data:

- Open Tableau
- Click Connect → Text File
- Load Housing_Data.csv
- Check data types (Price → Number, Date → Date, etc.)

Data Preparation:

- Remove null values
- Rename columns properly
- Create calculated fields:
- House Age = Year Sold – Year Built
- Years Since Renovation
- Price Bins
- **Create Key KPIs (Sheet 1):**
- Total Area (Sqft)
- Total Count of Houses
- Average Sale Price
- Use Text Table / KPI cards
- **Create Visualizations:**

Total Sales by Years Since Renovation

- Use Bar Chart
- Columns → Years Since Renovation
- Rows → Count of Sale Price

Distribution of House Age by Renovation Status

- Use Pie Chart
- Dimension → Renovation Status
- Measure → Count of Houses

House Age by Bedrooms, Bathrooms, Floors

- Use Grouped Bar Chart
- Columns → House Age
- Rows → Bedrooms/Bathrooms/Floors

Sale Price Distribution

- Use Histogram
- Create Bins for Sale Price

Apply Filters:

- Add filters for
- Bedrooms,Bathrooms,Floors
- Renovation Status Year

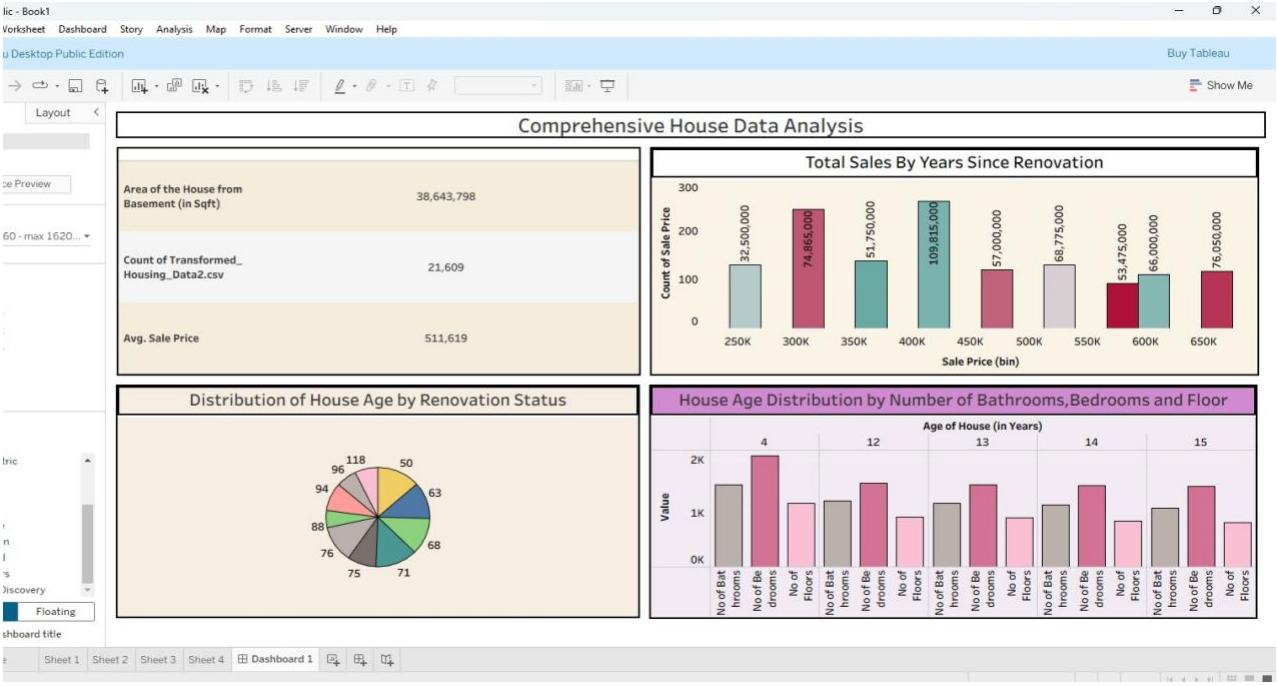
Create Dashboard:

- Click New Dashboard
- Set Size (Automatic / Desktop)
- Drag all sheets into dashboard
- Arrange properly (Tiled layout)
- Add Title: Comprehensive House Data Analysis

Formatting:

- Add colors for better readability
- Add borders & shading

Increase font size for KPIs, Add tooltips for insights



DashboardLink:

https://public.tableau.com/views/ComprehensiveHouseDataAnalysis_17715745169850/Dashboard1?:language=enUS&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Story:

- Story Point 1 – Introduction & KPI Overview
  Displays key metrics such as Total Houses Sold, Average Sale Price, Total Area (Sqft), and Renovation Count to provide a quick overview of the housing market performance.
- Story Point 2 – Sale Price Trend Analysis
  Analyzes sale price distribution and trends over the years, highlighting price growth patterns and fluctuations in the housing market.
- Story Point 3 – Renovation Impact Analysis
  Examines how renovation status and years since renovation influence house age and sale prices, helping understand value appreciation.
- Story Point 4 – Feature-Based Price Comparison
  Compares house prices based on number of bedrooms, bathrooms, floors, and area to identify which features significantly impact pricing.
- Story Point 5 – House Age & Market Distribution Analysis Explores the distribution of house age and its relationship with renovation status and structural features to identify market patterns.The story helps users understand housing market trends using a structured data narrative approach,enabling better real estate investment and business decisions.

Story Link:

https://public.tableau.com/views/HouseData-Story/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Dataset:

Dataset Link:

https://drive.google.com/file/d/1U0FLKgA1MaB_o5_1hqIF08KfdNGIO_xm/view?usp=drivesdk

Dataset	Records	Description
Transformed_Housing_Data2.csv	21,613	Transformed and cleaned housing dataset containing sale price, area, number of bedrooms, bathrooms, floors, renovation details, house age, and other property-related features used for housing market trend analysis and visualization in Tableau.
Area of House From Basement	38,643,798	Shows the total basement area across all houses, helping analyze property size distribution and its impact on housing market prices.
Average Sale Price	511,619	Shows the average house selling price, helping analyze overall market trends and compare prices based on property features.